

Bibliography Demo: Survival Analysis

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1 Introduction

The Cox proportional hazards model (Cox, 1972) is the most widely used method for survival analysis. The Kaplan-Meier estimator (Kaplan & Meier, 1958) provides a non-parametric estimate of the survival function.

Machine learning methods for survival analysis are reviewed in Hastie et al. (2009). Model discrimination is typically assessed using Harrell's concordance index (Harrell et al., 1982).

2 Methods

We follow the notation of Kalbfleisch and Prentice (2002). The Cox model specifies:

$$h(t \mid \mathbf{x}) = h_0(t) \exp(\mathbf{x}^\top \boldsymbol{\beta}) \quad (1)$$

Multiple citations: (Cox, 1972; Harrell et al., 1982; Kaplan & Meier, 1958).

3 Citation Commands

References

- Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society: Series B*, 34(2), 187–220. <https://doi.org/10.1111/j.2517-6161.1972.tb00899.x>
- Harrell, F. E., Califf, R. M., Pryor, D. B., Lee, K. L., & Rosati, R. A. (1982). Evaluating the yield of medical tests. *JAMA*, 247(18), 2543–2546. <https://doi.org/10.1001/jama.1982.03320430047030>

Table 1: Useful biblatex citation commands.

Command	Output example
<code>\textcite{cox1972}</code>	Cox (1972)
<code>\parencite{cox1972}</code>	(Cox, 1972)
<code>\citeauthor{cox1972}</code>	Cox
<code>\citeyear{cox1972}</code>	1972

- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning* (2nd). Springer. <https://doi.org/10.1007/978-0-387-84858-7>
- Kalbfleisch, J. D., & Prentice, R. L. (2002). *The statistical analysis of failure time data* (2nd). Wiley. <https://doi.org/10.1002/9781118032985>
- Kaplan, E. L., & Meier, P. (1958). Nonparametric estimation from incomplete observations. *Journal of the American Statistical Association*, 53(282), 457–481. <https://doi.org/10.1080/01621459.1958.10501452>